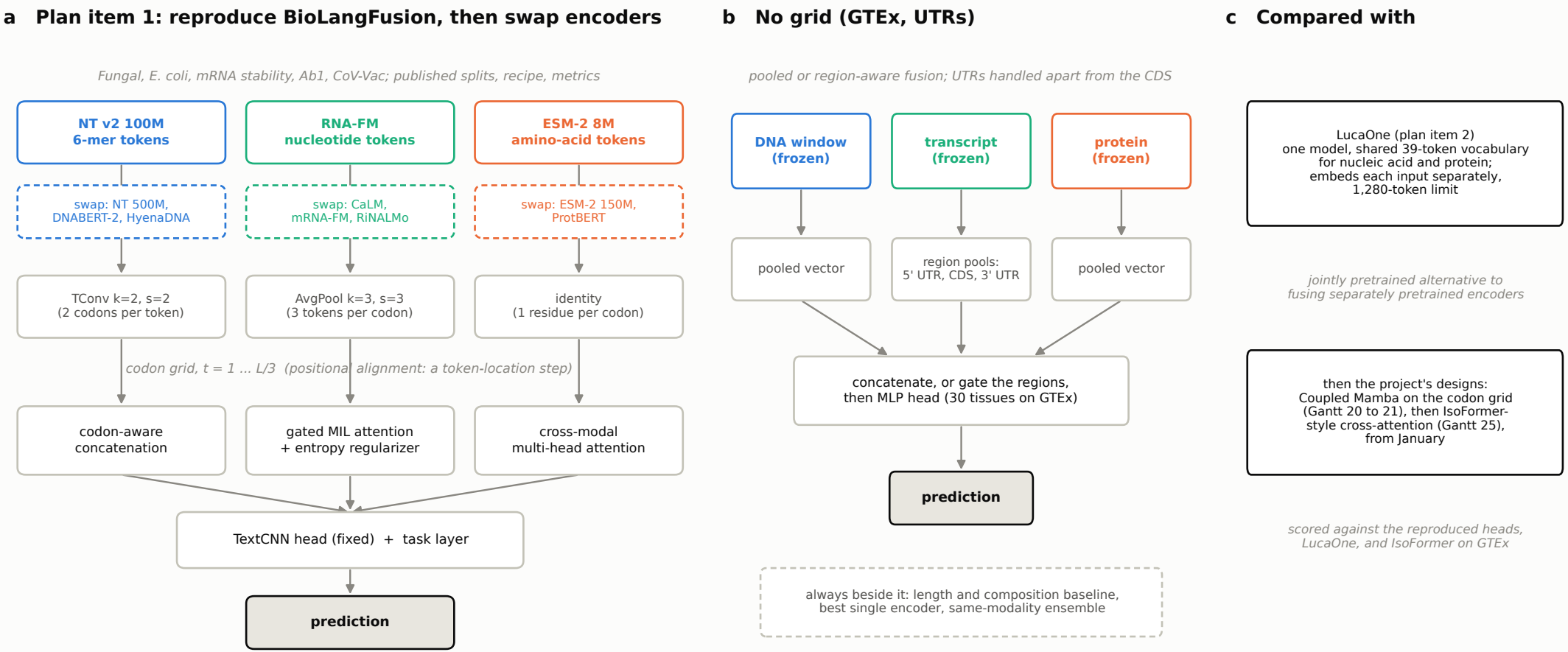
a Plan item 1: reproduce BioLangFusion, then swap encoders
Fungal, E. coli, mRNA stability, Ab1, CoV-Vac; published splits, recipe, metrics
NT v2 100M
6-mer tokens
RNA-FM
nucleotide tokens
ESM-2 8M
amino-acid tokens
swap: NT 500M,
DNABERT-2, HyenaDNA
swap: CaLM,
mRNA-FM, RiNALMo
swap: ESM-2 150M,
ProtBERT
TConv k=2, s=2
(2 codons per token)
AvgPool k=3, s=3
(3 tokens per codon)
identity
(1 residue per codon)
codon grid, t = 1 ... L/3 (positional alignment: a token-location step)
codon-aware
concatenation
gated MIL attention
+ entropy regularizer
cross-modal
multi-head attention
TextCNN head (fixed) + task layer
prediction
b No grid (GTEx, UTRs)
pooled or region-aware fusion; UTRs handled apart from the CDS
DNA window
(frozen)
transcript
(frozen)
protein
(frozen)
pooled vector
region pools:
5' UTR, CDS, 3' UTR
pooled vector
concatenate, or gate the regions,
then MLP head (30 tissues on GTEx)
prediction
always beside it: length and composition baseline,
best single encoder, same-modality ensemble
c Compared with
LucaOne (plan item 2)
one model, shared 39-token vocabulary
for nucleic acid and protein;
embeds each input separately,
1,280-token limit
jointly pretrained alternative to
fusing separately pretrained encoders
then the project's designs:
Coupled Mamba on the codon grid
(Gantt 20 to 21), then IsoFormer-
style cross-attention (Gantt 25),
from January
scored against the reproduced heads,
LucaOne, and IsoFormer on GTEx